

CASE 45

TOXIC SHOCK SYNDROME

Superantigens cause excessive stimulation of T cells and macrophages.

A conventional protein antigen activates only those T lymphocytes with a T-cell receptor specific for peptides derived from the antigen. This is typically a tiny subset of the total T-cell pool. For example, only about 1 in 10,000 circulating T cells from donors immunized with tetanus toxoid proliferate when subsequently stimulated by the toxoid. In contrast, a special class of antigens known as superantigens directly activates large numbers of T cells. Superantigens are bacterial and viral proteins that can bind, as whole unprocessed proteins, simultaneously to MHC class II molecules outside the peptide-binding groove and to certain $V\beta$ chains of the T-cell receptor. By engaging all T-cell receptors that share the same $V\beta$ chains rather than a receptor that recognizes the peptide in the MHC class II cleft, superantigens can activate a sizeable fraction of T cells and generate an amplified T-cell response far greater than that caused by the typical peptide antigen.

Superantigens first bind with high affinity to MHC class II molecules on antigen-presenting cells. As the concentration of bound superantigen increases, it will bind and cross-link T-cell receptors with the appropriate $V\beta$ specificity (Fig. 45.1). Cross-linking activates both the T cell and the antigen-presenting cell through signaling events downstream of the T-cell receptor and the MHC class II molecule, respectively. Because of the high affinity of superantigens for MHC class II molecules, a very small amount of superantigen results in intense T-cell signaling. For example, the toxic shock syndrome toxin-1 (TSST-1), which is produced by certain strains of the Gram-positive bacterium *Staphylococcus aureus*, stimulates all those T cells that express the $V\beta 2$ gene segment. As there is a limited repertoire of $V\beta$ gene segments, any superantigen will stimulate between 2% and 20% of all T cells.

TOPICS BEARING ON THIS CASE:

Cytokines

T-cell activation

Superantigens

Macrophage activation

Toxic shock

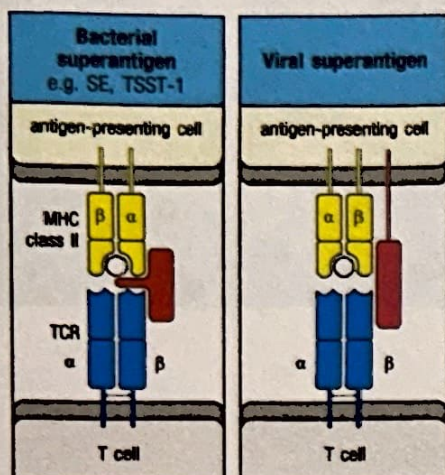


Fig. 45.1 Superantigens bind directly to T-cell receptors and MHC molecules. Superantigens interact with MHC class II molecules and T-cell receptors in a way that is quite distinct from the way in which normal peptide antigens bind. Superantigens bind independently to MHC class II molecules and to T-cell receptors, binding to the V β domain of the T-cell receptor, away from the peptide-binding site. The T-cell receptor α chain is not directly involved in binding superantigen.

This mode of stimulation is not specific for the pathogen and thus does not lead to adaptive immunity. Instead, it causes excessive production of cytokines by the large number of activated CD4 T cells, the predominant responding population. This massive release of cytokines has two effects on the host: systemic toxicity, which manifests itself as toxic shock syndrome; and suppression of the adaptive immune response. Both of these effects contribute to the pathogenicity of microbes that produce superantigens.

The case of Claire Bourbon: life-threatening shock from a superantigen.

Claire Bourbon was a healthy 16-year-old with a history of mild asthma and allergic rhinitis who suddenly became ill with a fever, general muscle aches, and dizziness. She felt nauseous and vomited. Claire's temperature rose to 39.8°C and her mother rushed her to the family physician. En route she briefly lost consciousness, and a red rash developed over her arms and spread rapidly to most of the body.

On arrival at the physician's she appeared quite ill and was immediately referred to the emergency department. She was alert but listless, and her general condition gave cause for concern. On examination, her temperature was 37.8°C, and heart rate and respiration rate were markedly elevated, at 140 beats per minute and 24 breaths per minute, respectively. Blood pressure was depressed—98/67 lying supine, 83/49 when seated, and 67/25 when standing—and showed evidence of significant volume depletion.

A bright red rash of flat and raised lesions was apparent on her trunk and extremities, but there were no petechiae (small subcutaneous hemorrhages) and no signs of localized infection.

Questioning revealed that Claire had not taken alcohol or drugs and had not been exposed to other ill individuals. Her last menstrual period had been 6 weeks before, and she had developed vaginal bleeding on the day previous to the onset of her illness. She had not used a tampon overnight, but had inserted one that morning, before she became ill.

Given Claire's critical status, extensive laboratory tests were conducted. Her white blood cell count was raised, at 21,000 cells μL^{-1} , with a predominance of neutrophils and band forms (immature neutrophils), indicating increased mobilization of neutrophils from the bone marrow. Serum electrolyte levels were within normal limits. The blood coagulation time was slightly prolonged and serum transaminase levels were raised; both of these signs are consistent with abnormal liver function.

Cerebrospinal fluid (CSF) was normal and showed no evidence of infection. Cultures of blood, urine, CSF, and vaginal fluid were made, and Claire was given a cephalosporin antibiotic (ceftriaxone) along with 2 liters of fluid intravenously. Her blood pressure improved and she was immediately admitted to the intensive care unit, where she developed petechiae. She was treated with intravenous fluids, two anti-staphylococcal antibiotics (oxacillin and clindamycin), cephalosporin (cefotaxime), and intravenous immunoglobulin (IVIG), and her overall condition gradually improved.

16-year-old female; systemic shock and bright red rash.

Started period day before.
Toxic shock syndrome?

Claire's blood, urine, and CSF cultures were sterile, but her vaginal culture was positive for abundant *S. aureus*. She was subsequently transferred to the regular in-patient ward and treated for 7 days with anti-staphylococcal antibiotics. The rash slowly faded.

Toxic shock syndrome.

Claire suffered from staphylococcal toxic shock syndrome (TSS), a striking example of the dramatic physiologic alterations caused by superantigens. TSS is a serious disease characterized by rapid onset of fever, a rash, organ failure, and shock. Most cases occur in menstruating women, typically in their teenage years, but cases do occur in all age groups. TSS is typically associated with a localized *S. aureus* infection (for example, subcutaneous abscesses, osteomyelitis, and infected wounds), staphylococcal food poisoning, or local colonization, as occurred in the vagina in this case. When kept in the vagina for a long time (more than 12 hours) tampons soaked with menstrual fluids can enhance the growth of the bacteria that are the source of superantigens. It is unlikely that the tampon played a part in Claire's illness because it was inserted less than 6 hours before the onset of symptoms. Toxigenic strains of *S. aureus* can produce enterotoxins (such as enterotoxin B) as well as TSST-1; these also act as superantigens with similar clinical consequences. In addition, microorganisms other than *S. aureus* secrete superantigens that can cause disease (Fig. 45.2).

Disease	Superantigen	TCR V β
Definite role for superantigen		
Toxic shock syndrome	TSST-1	V β 2
Staphylococcal food poisoning	SEA	V β 3, V β 11
	SEB	V β 3, V β 12, V β 14, V β 15, V β 17, V β 20
	SEC	V β 5, V β 12, V β 13.1-2, V β 14, V β 15, V β 17, V β 20
	SED	V β 5, V β 12
	SEE	V β 5.1, V β 6.1-3, V β 8, V β 18
Streptococcal toxic shock syndrome	SPE-A	V β 8, V β 12, V β 14
Scarlet fever	SPE-B	V β 2, V β 8
<i>Mycoplasma arthritidis</i> (rodent)	MAM	V β 17
<i>Clostridium perfringens</i>	Enterotoxin	V β 6.9, V β 22
Suspected role for superantigen		
HIV	CMV	V β 12
Type I diabetes mellitus	MMTV-like	V β 7
Rabies virus	Nucleocapsid	V β 8
Toxoplasmosis	?	V β 5
<i>Mycobacterium tuberculosis</i>	?	V β 8
<i>Yersinia enterocolitica</i>	?	V β 3, V β 6, V β 11
Kawasaki disease	?	V β 2, V β 8

Fig. 45.2 Superantigens, V β usage, and disease.

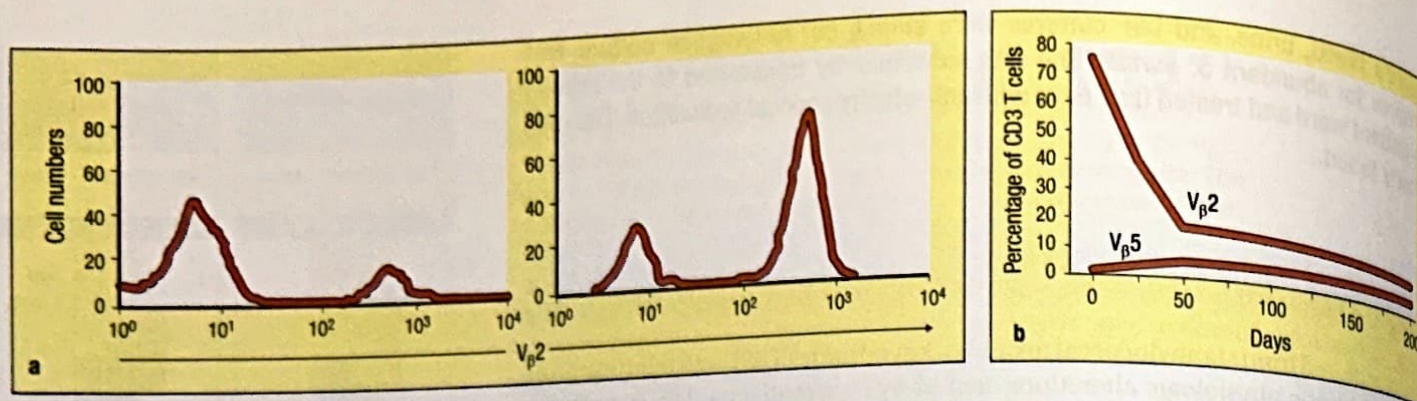


Fig. 45.3 Expansion in numbers of $V_{\beta}2$ T cells in toxic shock syndrome (TSS). Panel a, FACS analysis. Peripheral blood T cells from a normal control (left) and a patient with acute TSS (right). Cells are stained with anti- $V_{\beta}2$ monoclonal antibody with a

fluorescein tag. There is an increased percentage of $V_{\beta}2$ T cells in the patient. The horizontal axis represents the mean fluorescence intensity. Panel b, time course of persistence of high numbers and return to normal of $V_{\beta}2$ T cells in a patient with TSS.

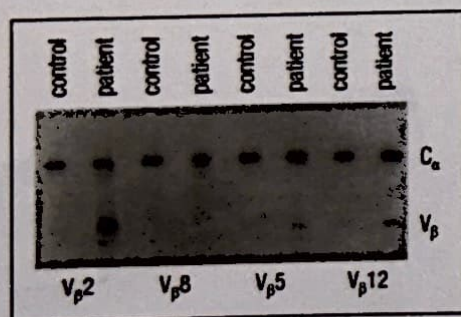


Fig. 45.4 RT-PCR analysis of T-cell receptor V_{β} mRNA. Autoradiograms of T-cell receptor chain transcripts amplified by reverse transcription followed by polymerase chain reaction (RT-PCR). T cells from a patient with toxic shock syndrome and a control individual were stimulated with anti-CD3 antibody and IL-2 before the extraction of RNA and generation of cDNA. Each reaction contained specific oligonucleotide primers to expand the particular V_{β} gene segment indicated (170–220 base pairs), as well as a C_{α} gene segment (about 600 base pairs) as a control to ascertain that equivalent amounts of mRNA were used. Photograph courtesy of Y. Choi.

Consistent with the $V_{\beta}2$ specificity of TSST-1, examination of the circulating lymphocytes from patients in the acute phase of TSS typically reveals a much higher proportion of circulating $V_{\beta}2$ T cells than cells using other V_{β} segments. As the illness resolves, there is a gradual return to near-normal proportions. The expansion in the numbers of $V_{\beta}2$ cells can be measured by examining the surface expression of $V_{\beta}2$ -containing T-cell receptors with the use of immunofluorescence (Fig. 45.3) or by semiquantitative measurement of mRNA transcripts encoding $V_{\beta}2$ T-cell receptor chains by using reverse transcription and the polymerase chain reaction (RT-PCR) (Fig. 45.4).

Although all the T cells activated by a given superantigen share a common V_{β} region, they will differ in their specificity for conventional peptide antigens. Sequencing of the T-cell receptors from superantigen-activated T cells reveals the use of different D and J gene segments by the β chains and a wide diversity of α chains. These receptors will encompass a wide variety of antigen specificities. In contrast, conventional antigens induce the clonotypic expansion of T cells—meaning that all the T cells in any given clone will have identical D and J gene segments in their β chains and identical α chains. Because the pool of V_{β} -restricted T cells activated by superantigen may contain autoreactive T cells, it has been postulated that superantigens could trigger autoimmune disease.

Many of the manifestations of TSS are the result of massive and unregulated cytokine production triggered by the activation of immune-system cells. TSST-1 is more effective than bacterial lipopolysaccharide in inducing the synthesis and secretion of interleukin (IL)-1 and tumor necrosis factor (TNF)- α by monocytes. TSST-1 is also a potent T-cell mitogen for those T cells whose receptors it engages; it also induces them to produce large amounts of cytokines, including IL-2 and interferon (IFN)- γ .

IL-1 and TNF- α are critical in the induction of the acute-phase response, characterized by fever and the production of IL-6. IL-1 and TNF- α also activate vascular endothelium and, together with IL-2, increase vascular permeability, with the subsequent leakage of fluid from the intravascular space into the perivascular space. These effects of the massive overproduction of TNF- α result in the toxic shock: edema and intravascular volume depletion leading to hypotension, shock, and multi-organ failure.

S. aureus is a common organism that colonizes 25–50% of the population, and nearly 50% of isolates produce superantigens. However, TSS is of rare occurrence. There are several reasons for this discrepancy. The staphylococcal cell wall contains peptidoglycans that bind to the Toll-like receptor TLR-2:TLR-6 on monocytes, resulting in activation of the NF κ B pathway (see Fig. 23.1). Although TLR signaling can cause a pro-inflammatory response, staphylococcal peptidoglycans induce an

anti-inflammatory response mediated by IL-10 production and subsequent apoptosis of antigen-presenting cells. This aborts the amplification of the inflammatory response associated with TSS. In addition, susceptibility to TSS correlates with a poor antibody response to TSST-1. Typically, individuals are exposed to bacterial numbers that are too low to produce dangerous levels of superantigen but are high enough to generate toxin-specific antibodies. But although most healthy individuals have protective antibody titers to TSST-1, more than 80% of patients with TSS lack anti-TSST-1 antibodies during the acute illness, and most fail to develop anti-TSST-1 antibodies after convalescence. Possible explanations include an individual's inability to mount an antibody response to TSST-1 and staphylococcal enterotoxins, or a specific inhibition of such a response by the toxins. In addition, certain HLA polymorphisms, particularly in the HLA-DQ allele, are associated with increased superantigen affinity for MHC class II molecules and more severe clinical symptoms. Because antibodies against staphylococcal and streptococcal superantigens confer protection against superantigen-mediated disease, and the vast majority of donors of the plasma pools used to prepare IVIG have antibodies against TSST, IVIG is a potential treatment for patients with TSS. IVIG should be used in conjunction with antibiotics, and its efficacy in neutralizing superantigens can differ between preparations.

Questions.

- 1 How do you determine whether a protein behaves as a superantigen?
- 2 Explain the rapid progression of clinical symptoms after the introduction of superantigen compared with the delay in apparent responses to conventional antigen.
- 3 What are the potential mechanisms of liver injury in TSS?
- 4 Is Claire susceptible to another bout of TSS?